

Abstract of Scientific Justification (300 words max) (2 points)

Supernovae (SNe) are some of the most interesting and studied phenomena in astronomy. The explosive death of a star can not only tell us about its life span and evolution but also the physical processes that happen during the final stages of a star's life. This is not only important in studying stellar evolution but also in other areas such as galaxy environments, studying the expansion of the universe, and so on.

Much of the science we know about supernovae comes from their light curves. By measuring how a supernova's brightness changes over time and using the constructed light curve, we can learn about the explosion, such as its decline rate, color evolution, and possible supernova type, given a well-sampled light curve.

For this project, we propose B, V, and, if possible, R band imaging of one to two recent supernovae with the Mont4K CCD on the 61" Kuiper Telescope. Supernovae cannot be chosen far in advance; any identified SN today will have faded by October and would have passed its peak brightness. Therefore, our targets may remain flexible: we will select targets from Transient Name Server reports in the days before the run, requiring $V < 19$, a declination between approximately -5 and $+40$ degrees, and enough angular separation from the moon. Our observations will be used to perform relative photometry with nearby comparison stars, and we will construct multi-band light curves for our sample of supernovae. These light curves will let us perform comparisons with known SN light curves, and depending on the phase we observe the supernova at and how well-sampled our light curve is, different science becomes possible, such as measuring peak brightness and color near maximum light or the slope of the radioactive decay tail after it.

Telescope Requested (0.5 points): 61" Kuiper Telescope

Instrument (0.5 points): Mont4K CCD, using primarily the B, V, and, if permitted, R filters.

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points):

Dark or grey time is preferred, since the targets are faint (approximately 18-19 mag). Bright time may be acceptable if the supernova is sufficiently bright and has a large enough separation from the moon, but a bright moon background will definitely increase the sky background and hence reduce the signal-to-noise ratio, especially for faint supernovae.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

Object Name	Right Ascension (h, m, s)	Declination (d, m, s)	V magnitude*	Object Type	Visibility
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AT 2026aawd	20:21:37.093	+23:41:09.61	18.01	Transient (SN candidate)	September - December
SN 2026aars	02:29:34.883	+23:06:58.44	18.42	SN	September - January

* These are the discovery magnitudes, as stated on the Transient Name Server.

Given the nature of transient science, any source discovered now would have changed significantly by the time of our observing night, whether that would be due to fading below the useful magnitude range or having passed its rise to peak brightness, which lasts around 10 to 20 days (e.g., for type Ia SN, see Riess et al. 1999; Hayden et al. 2010, and for type II see Ertini et al. 2026) and is one of the more interesting parts to study. Any new supernova candidate that satisfies the magnitude limit, declination, altitude, and moon-separation criteria can replace the currently listed targets.

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program, and what signal-to-noise is necessary for your observations? (2 points)

We require a signal-to-noise ratio of approximately 100 in each B, V, and R measurement. This is to constrain the shape of our light curves well enough to let us measure a rise or decline rate and compare against known SN light curves.

Using the simple approximation $SNR \sim \sqrt{Ft}$, the time required for $SNR = 100$ is $t \sim \frac{10,000}{F}$. To avoid dealing with zero points and getting accurate count rates and converting F into suitable units (for a published zero point for the Mont4K CCD, see Lim et al. 2015), let us make some simple assumptions. Past studies, such as the one in Xu et al. 2018, obtained $SNR = 300$ per 30 s frame on a 16.2 magnitude target with the Mont4K instrument. For the faint end of our observation, $V = 19$, and using the magnitude equation, we can estimate

$$t = 30s \times \left(\frac{100}{300}\right)^2 \times 10^{0.4(19-16.2)} \approx 45 s < 60s$$

Their observations used a broad white-light filter, whereas we are observing through individual broadband filters, which are considerably narrower. So to account for the fewer photons collected, we can make the rough estimate of needing 3 times the exposure time. This brings us to roughly 180s, and since these approximations ignore other factors such as the sky background, seeing factor, and the moon's brightness, we shall adopt a standard 240s exposure time for each of our filters. The final exposure times can be further refined by performing a few short tests on the night of the observations, by using a reference star, for example.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (2 points)

Our observations will give us brightness measurements of each supernova in multiple optical bands. Subtracting these bands from one another gives us the color at each epoch, so we can track how the supernova's color and brightness change over time. This, combined with additional observations, whether from later ASTR302 nights or from archival photometry of the sources, allows us to construct the supernova's light curve. More points let us constrain the shape of that curve, which is what lets us characterize its rise or decline rate and, given a well-sampled light curve, classify its type.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

If the weather doesn't allow us to get good data, we will use archival or publicly available data from telescopes such as the Zwicky Transient Facility (ZTF) to study supernova light curves. We would start by processing the data and move on to constructing a reasonable light curve, and finally, if possible, classify it using photometric classification packages, such as Superphot+ (De Soto et al. 2024), which was developed for fitting light curves from surveys such as ZTF.

References

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Appendix A: Target Visibility from Mt. Bigelow on 10/03/2026

Duration for which selected sources will be visible at the site of the telescope, prepared using Astroplan and the example codes given on the class GitHub. Our modified code takes in values from a .csv file downloaded from the transient name server, allowing us to find visible sources given the criteria listed in the previous sections. This way, new targets can be identified and observed promptly, even a few nights before the actual observation.

Fall 2026 ASTRONOMY 302 Observing Proposal

